

MOHIT VAIDYA

M.Sc. Physics (Electronics) | Computational Physics | Scientific Programming | Research & Instrumentation
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PROFESSIONAL SUMMARY

Physics graduate with a strong foundation in applied and computational physics, electronics, and precision instrumentation. Gained hands-on research and engineering experience at SAMEER — a national research institution — contributing to the fabrication and performance validation of medical-grade RF accelerating systems. Demonstrated initiative by independently developing Python-based scientific automation tools during this tenure. Currently deepening expertise in computational physics through applied simulation projects in Python and MATLAB, including quantum mechanical modelling and 3D finite difference thermal analysis.

CORE COMPETENCIES

Computational Physics & Numerical Methods • Python Scientific Computing • MATLAB Simulation & Modelling • Finite Difference Methods • Quantum Mechanics • RF & Microwave Systems • Vacuum Technology & Precision Instrumentation • Electronics & Circuit Design • Data Analysis & Visualisation • Scientific Documentation & Reporting

RESEARCH & WORK EXPERIENCE

Project Assistant — RF Systems & Instrumentation | SAMEER (Society for Applied Microwave Electronics Engineering and Research), Mumbai Oct 2024 – Apr 2025

National research institution under Ministry of Electronics & IT (MeitY). Contributed to design, fabrication, and validation of RF accelerating tubes for medical LINAC systems — high-precision scientific instrumentation for national healthcare infrastructure.

- Contributed to fabrication and characterisation of **RF accelerating structures for 6 MeV and 15 MeV medical LINAC systems** — precision scientific instruments used in cancer radiotherapy.
- Performed **vacuum integrity testing and helium leak detection** on accelerator tubes, ensuring compliance with stringent operational thresholds through systematic measurement and documentation.
- Conducted **RF performance measurements and validation** of microwave components including waveguides, RF couplers, and RF loads — measuring shunt impedance against design specifications using systematic deviation analysis.
- Independently developed a **Python automation tool for RF cavity data analysis** — replacing a manual MATLAB-based workflow with an automated pipeline for data ingestion, resonance peak detection (SciPy), impedance area integration, and publication-ready plot generation.
- Executed **precision cavity alignment and mechanical integration** of RF accelerating structures to tight dimensional tolerances; performed bake-out procedures and critical vacuum measurements with full QA traceability.
- Participated in **beam testing and full-system integration validation** of LINAC subsystems; collaborated with RF engineers, mechanical engineers, and testing personnel.

Independent Computational Physics Research | Self-directed

Apr 2025 – Present

Deepening applied expertise in computational physics and scientific programming through end-to-end simulation projects — each grounded in real physics problems, validated against analytical solutions, and documented to research standards.

- Advancing **Python proficiency for scientific computing** (NumPy, SciPy, Matplotlib) — building numerical solvers and simulation frameworks with rigorous validation and error quantification.
- Developing **MATLAB expertise for engineering and physics simulation** — 3D finite difference thermal modelling, gradient-based optimisation, and sparse matrix computation.
- Published two open-source computational physics projects on GitHub** with full physics derivations, validation reports, and structured documentation — demonstrating research-grade rigour and scientific communication.
- Preparing for **CSIR NET (July 2025)** — national eligibility test for Junior Research Fellowship and Lecturership in Physical Sciences.

COMPUTATIONAL PHYSICS PROJECTS

PsiEvolutionKit — Quantum Mechanics Simulation Toolkit | Python | github.com/v-mv/PsiEvolutionKit

- Built a 1D Schrödinger equation solver using **finite difference methods** — covering infinite square well, harmonic oscillator, and time-evolving wave packets with animated visualisations of quantum tunnelling, superposition, and energy levels.
- Achieved **<0.001% numerical error** vs. analytical solutions — demonstrating rigorous numerical validation and computational physics methodology.
- Documented with theory derivations, Jupyter notebooks, and customisation guides — structured to research publication standards.

3D Heatsink Thermal Modelling & Topology Optimisation | MATLAB | github.com/v-mv/3d-Heatsink

- Developed a **3D finite difference thermal solver** with a 7-point spatial stencil, Robin/Neumann boundary conditions — modelling real heat transfer physics in complex geometries.
- Applied **gradient-based topology optimisation (fmincon SQP)** to determine thermally optimal geometry; engineered sparse Laplacian matrix assembly reducing computational complexity from $O(N^2)$ to $O(N)$.
- Full physics derivation documented in PHYSICS.md — reflecting scientific rigour and technical communication discipline.

EDUCATION

Master of Science — Physics with Specialisation in Electronics | Mumbai University Sep 2021 – Jul 2023

GPA: 7.17/10 | Academic Topper — NES Ratnam College (2022–23)

- Core: Quantum Mechanics, Electromagnetic Theory, Applied Physics, Electronics, Numerical Methods, Solid State Physics

Bachelor of Science — Physics | Mumbai University

Jun 2018 – Jun 2021

GPA: 6.89/10 | Class Topper — IAPT NGPE Examination (Apr 2021)

TECHNICAL SKILLS

Scientific Programming: Python (NumPy, SciPy, Matplotlib), MATLAB (FDM solvers, optimisation, sparse matrices), Simulink

Electronics & Embedded: Embedded C, C++, Assembly, VHDL, Proteus, Keil µVision IDE

Instrumentation: RF Performance Testing, Vacuum Technology, Helium Leak Detection, Shunt Impedance Measurement

Research Tools: Jupyter Notebooks, GitHub, LaTeX-style documentation, QA & Technical Report Writing

AWARDS & RECOGNITION

- Academic Topper — M.Sc. Physics, NES Ratnam College of Arts, Science and Commerce (2022–23)
- Class Topper — National Graduate Physics Examination (NGPE), Indian Association of Physics Teachers (IAPT), 2021